

Delhi Public School

Subject: Biology | Grade: 12th | Class: 12th

Time: 2 Hours

Maximum Marks: 13

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

Objective Type Questions. Choose the correct option for each of the following questions. Each question carries 1 mark.
Time recommended: 15 minutes.

Q1. Which of the following is a classic example of codominance in humans? [Easy] [1]

- a. Skin color
- b. ABO blood grouping
- c. Height
- d. Eye color

Expected Answer: ABO blood grouping

Q2. In a cross between red-flowered (RR) and white-flowered (rr) *Mirabilis jalapa* plants, the F₁ plants are all pink (Rr). When F₁ plants are self-pollinated, what will be the phenotypic ratio in the F₂ generation? [Moderate] [1]

- a. 3 Red : 1 White
- b. 1 Red : 2 Pink : 1 White
- c. 9 Red : 3 Pink : 4 White
- d. All Pink

Expected Answer: 1 Red : 2 Pink : 1 White

Q3. In a stable population at Hardy-Weinberg equilibrium, the frequency of a recessive allele (q) is 0.4. What is the frequency of the heterozygous carrier genotype (2pq) in this population? [Moderate] [1]

- a. 0.16
- b. 0.36
- c. 0.48
- d. 0.24

Expected Answer: 0.48

Q4. The presence of homologous organs in different organisms, such as the forelimbs of whales, bats, and humans, indicates which type of evolution? [Easy] [1]

- a. Convergent evolution
- b. Divergent evolution
- c. Saltation
- d. Co-evolution

Expected Answer: Divergent evolution

Q5. What is the expected phenotypic ratio in the F₂ generation of a Mendelian dihybrid cross involving seed shape and seed color? [Moderate] [1]

- a. 3:1
- b. 1:2:1
- c. 9:3:3:1
- d. 1:1:1:1

Expected Answer: 9:3:3:1

Section B

Short Answer Questions. Answer the following questions in 30-50 words. Each question carries 2 marks.

Q6. What is a test cross? Explain its significance in determining the genotype of an organism showing a dominant phenotype. [Easy] [2]

Expected Answer: A test cross involves crossing an individual showing a dominant phenotype (but unknown genotype) with a homozygous recessive individual. If the offspring are 100% dominant, the parent was homozygous dominant. If there is a 1:1 ratio of dominant to recessive phenotypes, the parent was heterozygous. Its significance is to determine the unknown genotype of a dominant phenotyped plant/animal.

Q7. Differentiate between incomplete dominance and codominance, providing one biological example for each. [Moderate] [2]

Expected Answer: In incomplete dominance, the F1 hybrid exhibits an intermediate phenotype that is a blend of both parental traits (e.g., Pink flowers in Snapdragon/Mirabilis jalapa when red is crossed with white). In codominance, both alleles express themselves fully and independently in the F1 hybrid (e.g., IA and IB alleles in human AB blood group expressing both A and B antigens).

Q8. State the Hardy-Weinberg principle. List any three factors that are known to disrupt Hardy-Weinberg equilibrium in a natural population. [Hard] [2]

Expected Answer: The Hardy-Weinberg principle states that allele frequencies in a population are stable and remain constant from generation to generation in the absence of evolutionary disturbing factors. Factors that disrupt this equilibrium are: 1. Gene migration/Gene flow, 2. Genetic drift, 3. Mutation, 4. Genetic recombination, 5. Natural selection.

Section C

Long Answer Questions. Answer the following detailed questions. Each question carries 1 mark. Time recommended: 20 minutes.

Q9. Using the phenomenon of Industrial Melanism observed in England (moths before and after industrialization), explain Charles Darwin's theory of Natural Selection. How does this serve as an observable example of evolutionary change? [Moderate] [1]

Expected Answer: Before industrialization in England, light-colored lichens covered tree trunks, camouflaging white-winged moths (Biston betularia) from predators, while dark-winged (melanic) moths were easily spotted and eaten. Post-industrialization, coal soot killed lichens and blackened tree trunks. Now, dark moths had a survival advantage due to camouflage, whereas white moths were preyed upon. This shifted the population frequency in favor of melanic moths. This supports Darwin's natural selection theory: nature selects the variants that are best adapted to the changing environment, leading to differential survival and reproductive success.

Q10. State Mendel's Law of Independent Assortment. Demonstrate this law using a dihybrid cross between a homozygous round yellow-seeded pea plant (RRYY) and a wrinkled green-seeded pea plant (rryy). Show the gametes, F1 generation, and the F2 generation punnett square representation. [Hard] [1]

Expected Answer: Mendel's Law of Independent Assortment states that when two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters during gamete formation. - Parents: RRYY (Round Yellow) x rryy (Wrinkled Green). - Gametes: RY and ry. - F1 Generation: RrYy (Round Yellow). - F1 selfing (RrYy x RrYy) yields 4 gamete types: RY, Ry, rY, ry. - F2 Punnett Square reveals a 16-box matrix with phenotypic ratio 9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green, which demonstrates that seed shape and seed color sort completely independently.

— End of Question Paper —